Pre-work · The Peripheral and Autonomic Nervous Systems

The Peripheral Nervous System, the Cranial Nerves, the Nerve Plexuses, and the Autonomic Nervous System. Read all four Part A chapters first. This is the largest packet in the module because it is four chapters, and the twelve cranial nerves are learned by mnemonic, so build yours in Panel 2 and use it all week.

Module 5 · 4 of 5 | Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

3 min

PANEL 1 OF 3

Turn the four peripheral chapters into mini visuals. Eight chunks. The chains carry the pathways, the grids carry the two lists you have to memorize, and the split is the pair that the whole autonomic chapter is built on.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 **LADDER**

A nerve, from axon to whole nerve

Three rungs, single axon at the bottom, fascicle in the middle, whole nerve at the top. Write the connective tissue wrapping beside each rung, and note which axons inside are myelinated and which are not. The names give away the level each one wraps, so write that out beside the ladder.

2 **CHAIN**

From rootlet to ramus

One chain with arrows. Rootlets, posterior root with its ganglion and anterior root, then the mixed spinal nerve at the intervertebral foramen, then the split into posterior ramus, anterior ramus, meningeal branch, and rami communicantes. Write what each of the four branches supplies at the end of the chain.

3 **GRID**

The twelve cranial nerves

Number and name down the side, I through XII. Class as sensory, motor, or mixed, primary function, and skull foramen across the top. Then circle the three foramina that carry more than one nerve and bracket the nerves that share each one. Write both memory sentences down the margin before you leave this frame.

4 **HUB**

The sensory receptors, drawn where they live

A block of skin at the hub, epidermis down to hypodermis, with muscle and tendon beside it. Place each named receptor at its true depth: Merkel discs in the deepest epidermis, Meissner corpuscles in the dermal papillae, free nerve endings and hair follicle receptors through the dermis, Pacinian corpuscles deep, Ruffini corpuscles deep, then muscle spindles, tendon organs, and joint receptors. Write what each one detects beside it.

5 **GRID**

The five plexuses

Plexus down the side, cervical through coccygeal. Formed by which anterior rami, region served, and the deficit if it is damaged across the top. Then write the one exception underneath: the anterior rami that form no plexus at all, and where they go instead.

6 **SPLIT**

Somatic against autonomic

One split. Effector, voluntary or involuntary, number of motor neurons in the pathway, effect on the effector, and whether you are aware of it. Five rows on each branch. The row that names the number of neurons is the one everything else follows from, so put it first.

7 **GRID**

Sympathetic against parasympathetic

Feature down the side, sympathetic and parasympathetic across the top. Other name, origin in the CNS, location of the ganglia, overall role, and four sample effects on named organs. The two other names are a map of the origins, so write them at the top and check the origin row against them.

8 **CHAIN**

The two-neuron autonomic pathway

Four boxes with arrows: preganglionic neuron, autonomic ganglion, postganglionic neuron, effector tissue. Mark which axon is myelinated and which is not. Then draw the chain twice, once with the ganglion near the cord and once with it at the organ, and label which division each version belongs to.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

3 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

DRAWING 1

A spinal nerve, from the cord to its rami

BRAIN DUMP FIRST

Two minutes. Both roots, the ganglion, the mixed nerve, and every branch off it.

Draw a cord cross-section inside a vertebra. Bring rootlets off the posterior and anterior surfaces, gather them into the two roots, put the posterior root ganglion on the sensory root, and join the roots into the mixed spinal nerve at the intervertebral foramen. Then split the nerve into the posterior ramus going to the back, the anterior ramus going around the trunk, the meningeal branch turning back into the canal, and the rami communicantes running to a sympathetic trunk ganglion. Color sensory and motor axons differently the whole way.

In your notebook, head the page **M5-10 A spinal nerve, from the cord to its rami** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can start at a receptor in the skin of the back and follow one continuous line into the cord on your own drawing, naming every structure you cross.

DRAWING 2

The twelve cranial nerves on the base of the brain, and your own mnemonic

BRAIN DUMP FIRST

Two minutes. All twelve, in order, attached where they belong.

Draw the inferior surface of the brain: frontal lobes at the front, temporal lobes to the sides, the three brainstem regions down the middle, and the cerebellum behind. Attach all twelve nerves in order and number them, with I and II reaching the cerebrum and diencephalon and III through XII on the brainstem. Then write out the two memory sentences from Part A, one for the names and one for the sensory, motor, or mixed pattern. Now build your own for each, using words that mean something to you, and write both under your drawing.

In your notebook, head the page **M5-11 The twelve cranial nerves on the base of the brain, and your own mnemonic** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can recite your own mnemonic and expand every word into the correct nerve name and class, with the drawing covered.

DRAWING 3

The two autonomic divisions, side by side

BRAIN DUMP FIRST

Two minutes. Where each one leaves the CNS, where its ganglia sit, and how far each neuron runs.

Draw the brain and spinal cord twice, side by side. On the sympathetic side mark the T1 to L2 lateral horns as the origin, draw the sympathetic trunk as a chain of ganglia beside the vertebral column, add the prevertebral ganglia near the abdominal arteries, and run a short preganglionic and a long postganglionic axon out to an organ. On the parasympathetic side mark the nuclei of CN III, VII, IX, and X and the S2 to S4 segments, and run a long preganglionic axon to a terminal ganglion at the organ with a short postganglionic axon beyond it.

In your notebook, head the page **M5-12 The two autonomic divisions, side by side** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can look at either drawing and say which axon is long, which is short, and why the ganglion position decides it.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A tumor grows at the jugular foramen on the right. The patient has difficulty swallowing, a hoarse voice, and weakness turning the head and shrugging the right shoulder. Tongue movement and facial expression are normal. Name the nerves involved.

Answer. Cranial nerves IX, X, and XI.

Reasoning. Work from the opening, because a foramen is a place where several nerves are squeezed together and injured as a set. The jugular foramen carries the glossopharyngeal, vagus, and accessory nerves side by side, and the three deficits described match those three jobs exactly: swallowing and taste from the back of the tongue for IX, voice and swallowing for X, and the sternocleidomastoid and trapezius for XI. Now check what is spared, because that is what rules out the neighbours. Tongue movement is normal, so the hypoglossal nerve is intact, and it leaves by a different opening, the hypoglossal canal. Facial expression is normal, so the facial nerve is intact, and it travels the internal acoustic meatus instead. Learn the foramina and the clusters of signs come with them.

Set A · Name it

1. The connective tissue wrapping around a single axon within a nerve is the ______________________.
2. The nerve arising from the cervical plexus that supplies the diaphragm is the ______________________.
3. The receptor of the deep dermis that detects deep pressure and vibration is the ______________________.
4. The parasympathetic ganglia, located near or within the wall of the target organ, are the ______________________.
5. Write a mnemonic of your own for the twelve cranial nerve names in order, and a second one for the sensory, motor, or mixed pattern.

Set B · Predict it

1. A patient loses movement of the sternocleidomastoid and trapezius on one side. Name the nerve and predict the other two nerves at risk with it.
2. An injury damages the brachial plexus on the left. Predict the region affected and explain why damage to a single spinal nerve root would be less complete.
3. A patient is startled by a loud noise. Predict what the sympathetic division does to the pupils, the heart, the airways, and digestion.
4. The vagus nerve is cut low in the neck. Predict which organs lose parasympathetic supply and explain how one nerve reaches so many.

Set C · Tell it apart

1. Distinguish the somatic from the autonomic motor pathway by the number of neurons, the effect on the effector, and awareness.
2. Compare where the sympathetic and parasympathetic ganglia sit, and explain how that difference changes the length of each axon.
3. Distinguish the myenteric from the submucosal plexus by position in the gut wall and by what each one controls.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet

2 min

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

M5-

A nerve, from axon to whole nerve

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

M5-2

From rootlet to ramus

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

M5-3

The twelve cranial nerves

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

M5-4

The sensory receptors, drawn where they live

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M5-5

The five plexuses

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

M5-6

Somatic against autonomic

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

M5-7

Sympathetic against parasympathetic

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

M5-8

The two-neuron autonomic pathway

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.